**D20A / 37**

## Experiment No: 4

**AIM:** To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

**THEORY:**

Blockchain is a **distributed and tamper-resistant digital ledger** used to record transactions and data. In this experiment, blockchain technology is used to implement a **Digital Warranty and Product Ownership Transfer System**.

The system uses a **Solidity smart contract** deployed on the **Ethereum Sepolia Testnet** through **Remix IDE**. MetaMask acts as the blockchain wallet and is used to connect the user's account with Remix and authorize transactions.

The smart contract stores product information such as:

- Product ID
- Product name
- Owner's blockchain address
- Warranty period
- Purchase date
- Product existence status

The contract also provides functions to register a product, check its warranty status, retrieve product information, and transfer ownership.

A major blockchain feature demonstrated in this project is **access control through the smart contract**. The `transferOwnership()` function contains a condition that allows only the current owner to transfer the product:

```
require(
   products[_productId].owner == msg.sender,
   "Only current owner can transfer"
);
```

Therefore, when Account 1 attempted to transfer the product after ownership had already been transferred to Account 2, the transaction was rejected with:

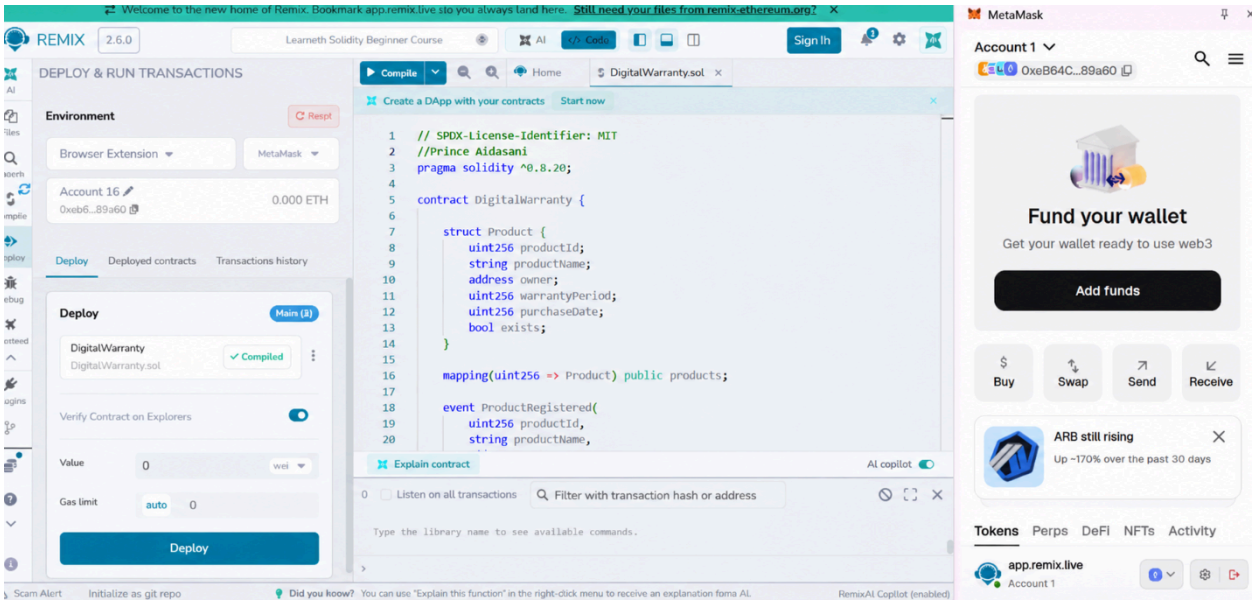
Only current owner can transfer

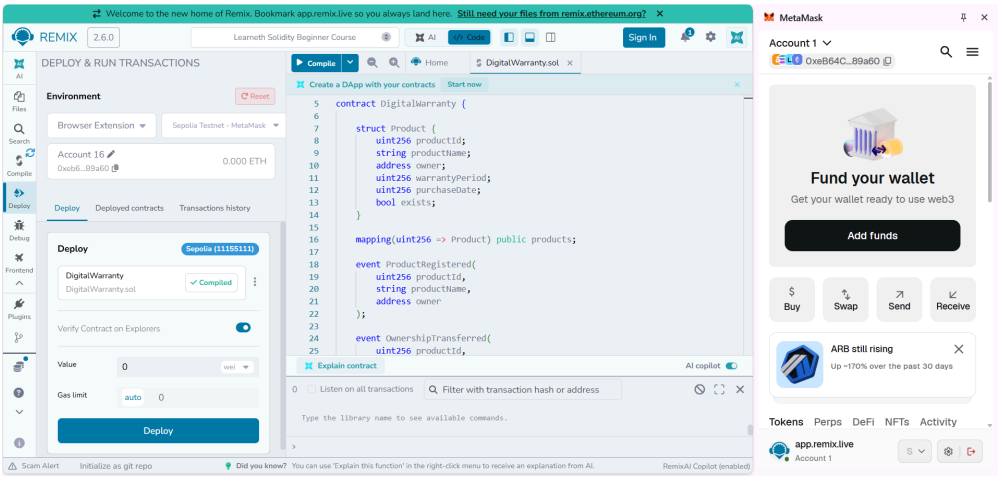
This demonstrates that the smart contract automatically enforces the ownership rule.

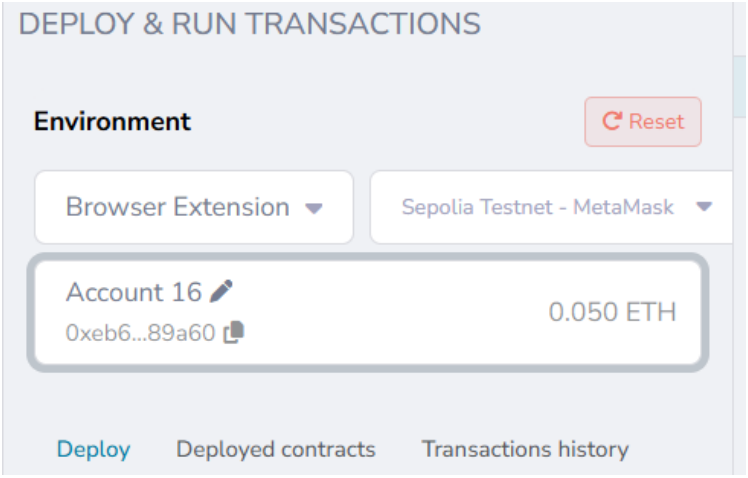
The experiment follows the lab requirements of setting up MetaMask, connecting a testnet to Remix, creating and deploying a Solidity smart contract, checking transaction details, and interacting with the deployed contract.

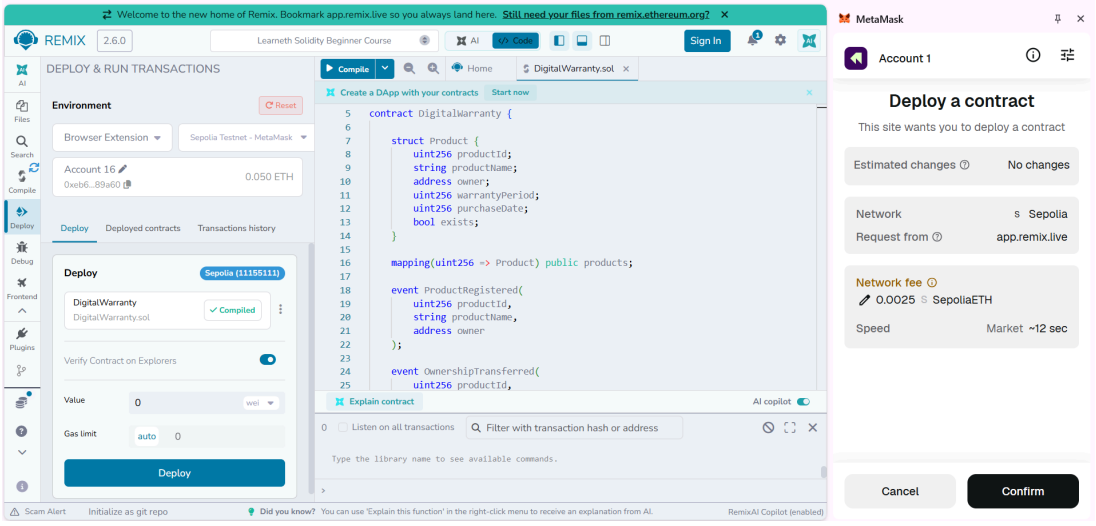
**STEPS and OUTPUT:**

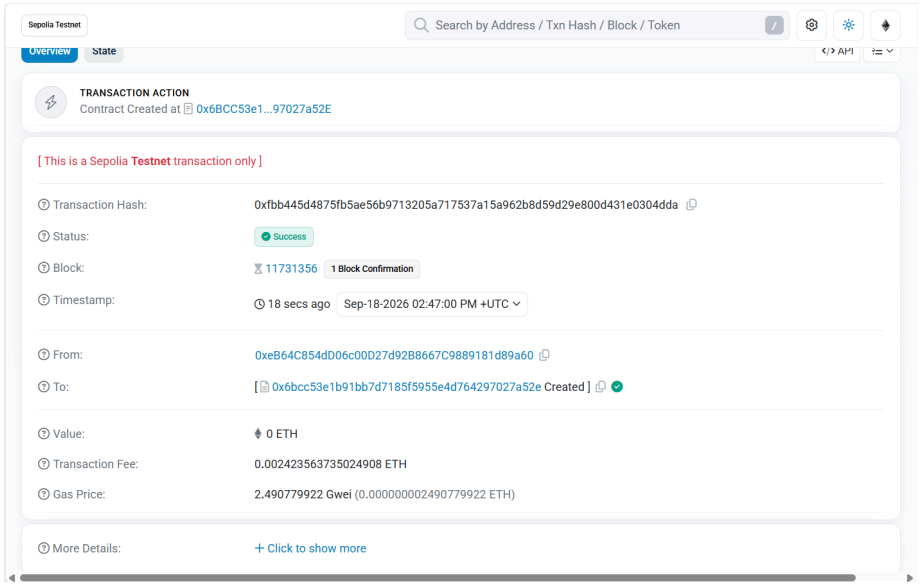
1. Setup metamask and connect with Remix IDE
2. Create the solidity smart contract
3. Compile and deploy it.
4. Register a product
5. Verify product details
6. Check warranty
7. Transfer product ownership
8. Verify new ownership
9. Demonstrate unauthorized transfer restriction

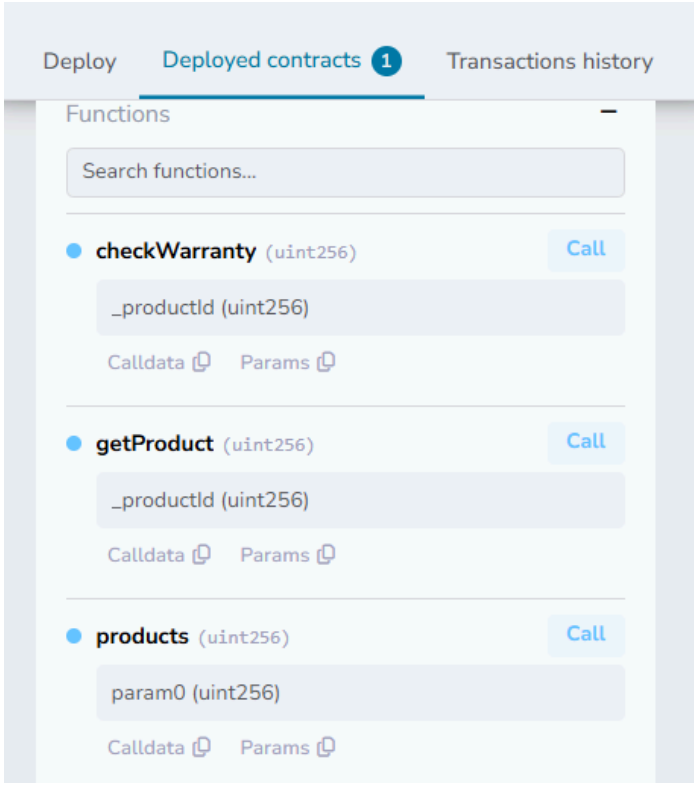


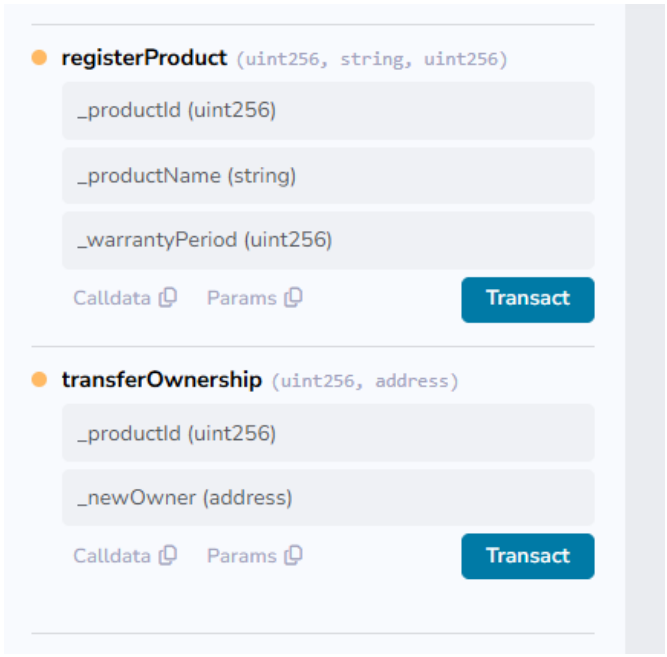

Welcome to the new home of Remix. Bookmark app.remix.live so you always land here. Still need your files from remix.ethereum.org?
REMIX 2.6.0
Learneth Solidity Beginner Course
Sign In
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
Sepolia Testnet - MetaMask
Account 16
0xeb6...89a60
0.000 ETH
Deploy
Deployed contracts
Transactions history
DigitalWarranty
DigitalWarranty.sol
Compiled
Verify Contract on Explorers
Value
Gas limit
auto
Deploy
DigitalWarranty.sol
contract DigitalWarranty {
struct Product {
uint256 productId;
string productName;
address owner;
uint256 warrantyPeriod;
uint256 purchaseDate;
bool exists;
}
mapping(uint256 => Product) public products;
event ProductRegistered(
uint256 productId,
string productName,
address owner
);
event OwnershipTransferred(
uint256 productId,
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or address
Type the library name to see available commands.
Scam Alert
Initialize as git repo
Did you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI.
RemixAI Copilot (enabled)
MetaMask
Account 1
0xeB64C...89a60
Fund your wallet
Get your wallet ready to use web3
Add funds
Buy
Swap
Send
Receive
ARB still rising
Up ~170% over the past 30 days
Tokens
Perps
DeFi
NFTs
Activity
app.remix.live
Account 1

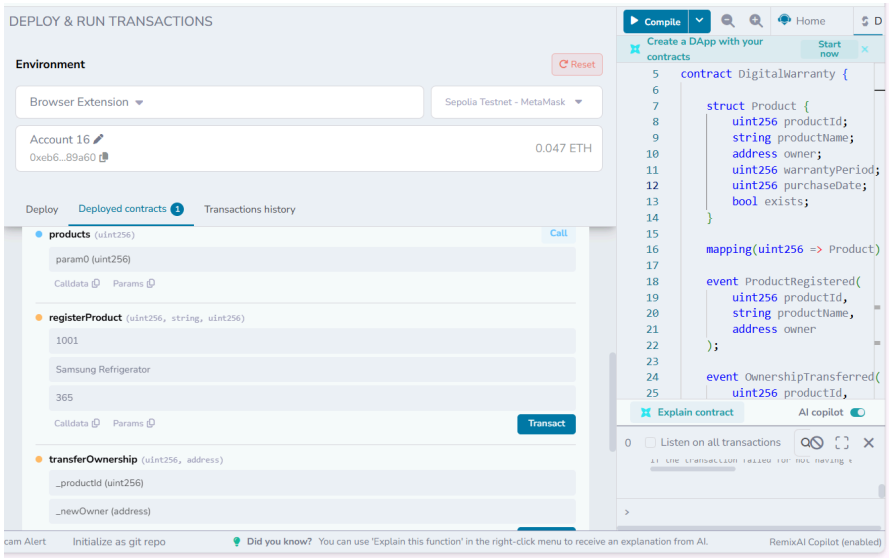

DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
Sepolia Testnet - MetaMask
Account 16
0xeb6...89a60
0.050 ETH
Deploy
Deployed contracts
Transactions history

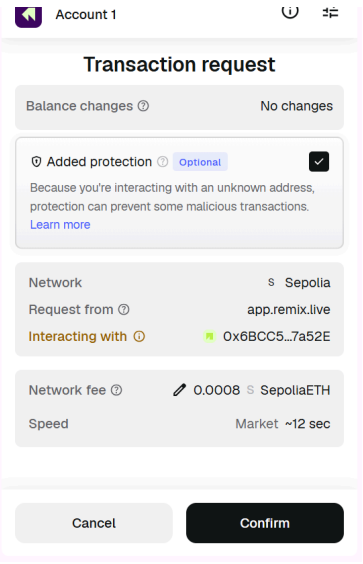

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Filter with transaction hash or address
Type the library name to see available commands.
Scam Alert
Initialize as git repo
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RemixAI Copilot (enabled)
MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
app.remix.live
Network fee
0.0025 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


Sepolia Testnet
Search by Address / Txn Hash / Block / Token
Overview
State
TRANSACTION ACTION
Contract Created at 0x6BCC53e1...97027a52E
[ This is a Sepolia Testnet transaction only ]
Transaction Hash:
0xfbb445d4875fb5ae56b9713205a717537a15a962b8d59d29e800d431e0304dda
Status:
Success
Block:
11731356
1 Block Confirmation
Timestamp:
18 secs ago
Sep-18-2026 02:47:00 PM +UTC
From:
0xeB64C854dD06c00D27d92B8667C9889181d89a60
To:
[ 0x6bcc53e1b91bb7d7185f5955e4d764297027a52e Created ]
Value:
0 ETH
Transaction Fee:
0.002423563735024908 ETH
Gas Price:
2.490779922 Gwei (0.000000002490779922 ETH)
More Details:
+ Click to show more


Deploy
Deployed contracts 1
Transactions history
Functions
Search functions...
checkWarranty (uint256)
Call
_productId (uint256)
Calldata
Params
getProduct (uint256)
Call
_productId (uint256)
Calldata
Params
products (uint256)
Call
param0 (uint256)
Calldata
Params


registerProduct (uint256, string, uint256)
_productId (uint256)
_productName (string)
_warrantyPeriod (uint256)
Calldata
Params
Transact
transferOwnership (uint256, address)
_productId (uint256)
_newOwner (address)
Calldata
Params
Transact


DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
Sepolia Testnet - MetaMask
Account 16
0xeb6...89a60
0.047 ETH
Deploy
Deployed contracts 1
Transactions history
products (uint256)
Call
param0 (uint256)
Calldata
Params
registerProduct (uint256, string, uint256)
1001
Samsung Refrigerator
365
Calldata
Params
Transact
transferOwnership (uint256, address)
_productId (uint256)
_newOwner (address)
Compile
Home
Create a DApp with your contracts
Start now
contract DigitalWarranty {
struct Product {
uint256 productId;
string productName;
address owner;
uint256 warrantyPeriod;
uint256 purchaseDate;
bool exists;
}
mapping(uint256 => Product)
event ProductRegistered(
uint256 productId,
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Explain contract
AI copilot
Listen on all transactions
Initialize as git repo
Did you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI.
RemixAI Copilot (enabled)


Account 1
Transaction request
Balance changes
No changes
Added protection
Optional
Because you're interacting with an unknown address, protection can prevent some malicious transactions.
Learn more
Network
Sepolia
Request from
app.remix.live
Interacting with
0x6BCC5...7a52E
Network fee
0.0008 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm

**getProduct** (uint256) Call

1001

Calldata Params

0:uint256: 1001

1:string: Samsung Refrigerator

2:address: 0xeB64C854dD06c00D27d92B8667C9889181d89a60

3:uint256: 365

4:uint256: 1789743204

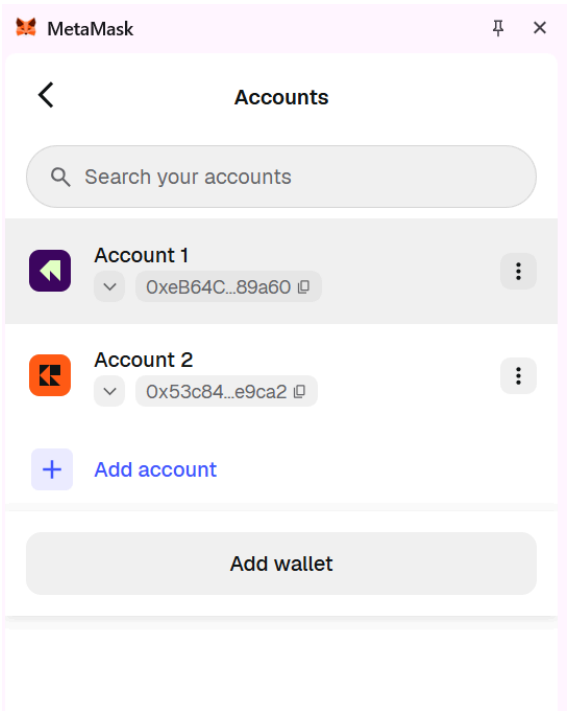
5:bool: true

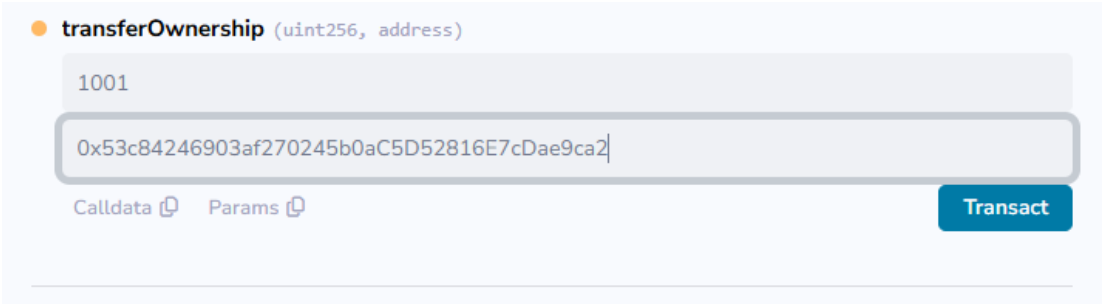
**checkWarranty** (uint256) Call

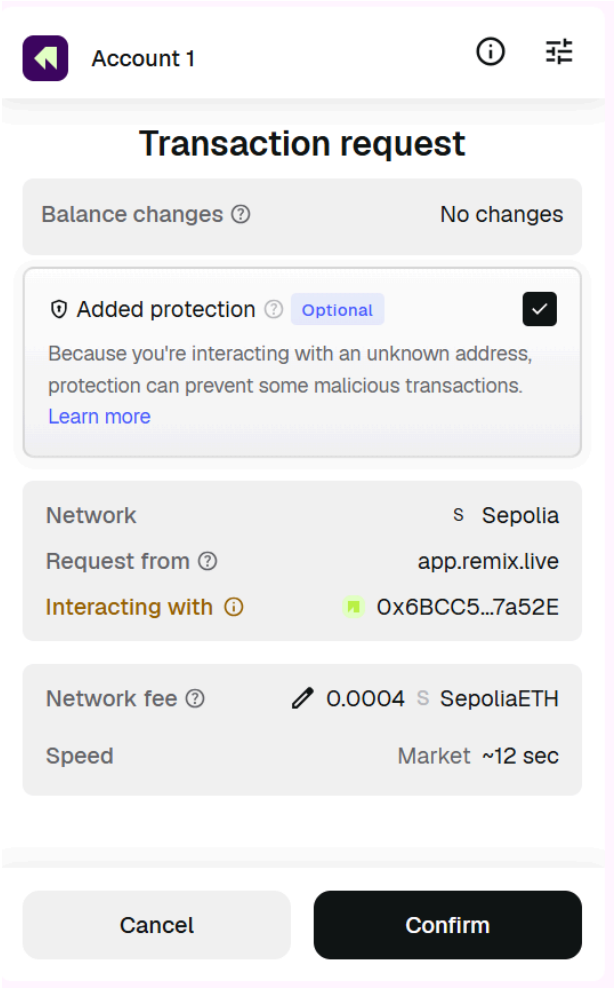
1001

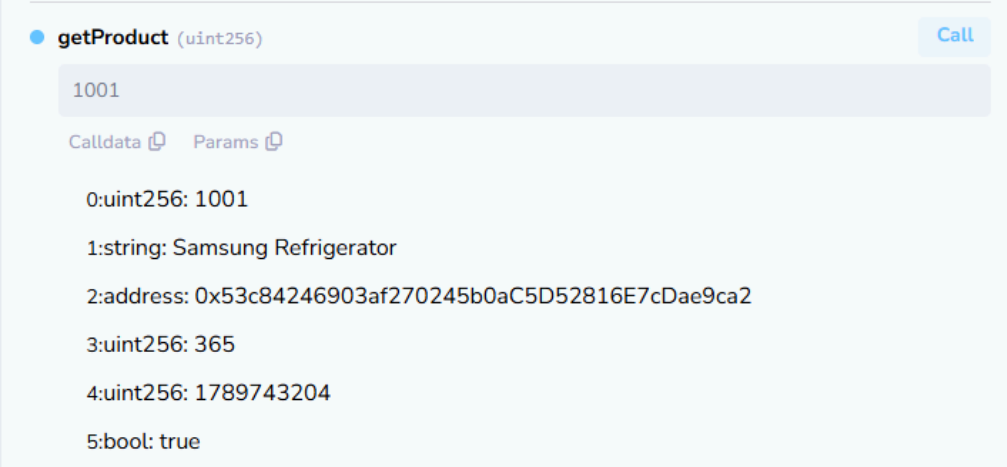
Calldata Params

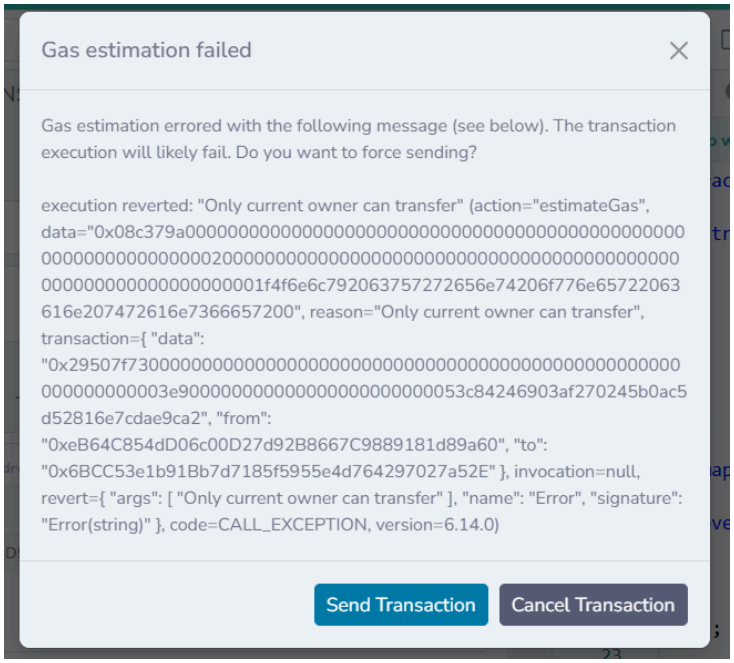
0:bool: true




transferOwnership (uint256, address)
1001
0x53c84246903af270245b0aC5D52816E7cDae9ca2
Calldata
Params
Transact


Account 1
Transaction request
Balance changes
No changes
Added protection
Optional
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Network
Sepolia
Request from
app.remix.live
Interacting with
0x6BCC5...7a52E
Network fee
0.0004 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm




**Conclusion:**

The blockchain-powered Digital Warranty and Product Ownership Transfer System was successfully implemented using Solidity and Remix IDE. The smart contract was deployed on the Sepolia Testnet using MetaMask. A product was registered with its warranty details, the warranty status was verified, and ownership was successfully transferred to another blockchain account. The updated ownership and active warranty were verified through the deployed smart contract.